

Project Title: Cyber Threat Intelligence Dashboard

Internship Domain: Cyber Security

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Organization: Elevate Labs

ThreatNexus AI

Introduction

ThreatNexus AI is an AI-powered Cyber Threat Intelligence (CTI) web application designed to monitor, analyze, and visualize cybersecurity threats through an interactive Security Operations Center (SOC) dashboard. The platform combines cybersecurity concepts with modern web technologies to help users understand global cyber threats in an intuitive way. It enables users to monitor live threat activities, perform VirusTotal IOC (Indicator of Compromise) lookups, explore interactive dashboards, and generate downloadable security reports. The project was developed to demonstrate practical cybersecurity analysis, data visualization, and full-stack web development skills.

Abstract

The main objective of this project is to build a centralized platform that provides meaningful insights into cyber threats using simulated threat intelligence data. ThreatNexus AI analyzes nearly 1,500 cybersecurity records containing information such as IP addresses, countries, attack types, severity levels, timestamps, and threat status. The application calculates AI-based threat scores, identifies risk levels, displays attack patterns through interactive charts, and provides security recommendations. Additionally, the platform integrates the VirusTotal API to perform real-time IP and domain reputation lookups. Features such as live threat monitoring, global threat visualization, analytics dashboards, and downloadable reports make the system suitable for educational, portfolio, and cybersecurity demonstration purposes.

Tools Used

- Python
- Flask
- HTML5
- CSS3
- JavaScript
- Plotly
- Pandas
- SQLite
- SQLAlchemy
- VirusTotal API
- ReportLab
- Git & GitHub

Steps Involved in Building the Project

1. Planned the overall architecture and workflow of the Cyber Threat Intelligence platform.
2. Designed responsive user interfaces using HTML5, CSS3, and JavaScript.
3. Developed the backend using Python and the Flask framework.
4. Created a realistic dataset containing approximately 1,500 simulated cybersecurity threat records.
5. Integrated the VirusTotal API to perform IOC lookups for IP addresses and domain names.
6. Implemented AI-based threat analysis to calculate threat scores, identify risk levels, determine top attack types, and generate security recommendations.
7. Built interactive dashboards including a Global Threat Map, Threat Timeline, Severity Distribution, Attack Type Analysis, and Live Threat Feed using Plotly.
8. Added search, filtering, analytics, and automated report generation in CSV and PDF formats.
9. Performed testing, debugging, and optimization before publishing the complete project on GitHub.

Conclusion

ThreatNexus AI successfully demonstrates the practical implementation of Cyber Threat Intelligence concepts using Flask, Python, and interactive data visualization techniques. Developing this project enhanced my understanding of cybersecurity analytics, API integration, dashboard development, and full-stack web application design. It also improved my problem-solving abilities and provided hands-on experience in building a professional, portfolio-ready cybersecurity application suitable for internships and technical interviews.